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Technical Workshop 2: Code-based life cycle assessment at NRL: introduction to LiAISON, CE ABM, and other tools.

Julien Walzberg, Tapajyoti Ghosh

Part 2: Circular Economy Agent-Based Model

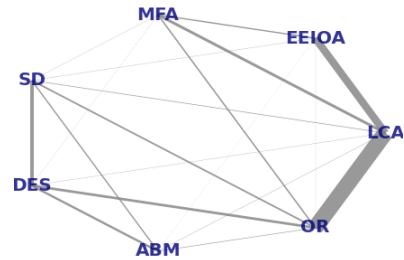
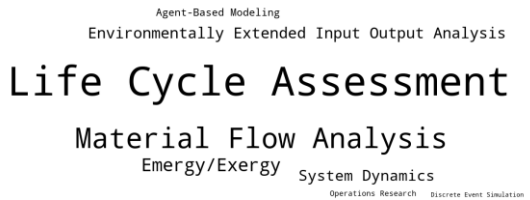
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Background – Problem statement

- A techno-economic representation of a product system is necessary but not always sufficient.
- Modeling detailed human or organization behaviors can improve LCA realism. For example:
 - Transitioning to a circular economy (CE) implies changes in production and consumption behaviors → a better modeling of those behaviors can help inform when CE systems are truly beneficial.
 - Information and communication technologies (ICT) are notoriously hard to model in LCA because of the uncertainty around user behavior → once again, incorporating information on behavior can provide better insights into the system and its environmental impacts.
- From recent advances in economics (e.g., from Khaneman and Thaler):
 - Human behaviors (and therefore organizational behaviors) are **not necessarily rational** (i.e., do not necessarily maximize utility).
 - They are **heterogeneous, constrained** (by the technological environment, by social norms) & **evolve**.

“[...] homo economicus can think like Albert Einstein, store as much memory as IBM’s Big Blue and exercise the willpower of Mahatma Gandhi. [...] But [...] real people have trouble with long division [...], sometimes forget their spouse’s birthday, and have a hangover on New Year’s Day.” – Richard H. Thaler & Cass R. Sunstein

Background – Methodology choice



- Methods from industrial ecology have been mostly used for circularity assessment (see word cloud).
- Combining methods from industrial ecology and complex systems science (graph) could alleviate some of their respective shortcomings.

ABM	What are the interactions among a systems' individual parts and how do they drive its overall behavior?	1. Models heterogeneity (system structure is not prescribed) 2. Represents social interactions 3. Models decisions that are not necessarily rational 4. Information on parts and whole of the system 5. Includes feedback loops 6. Dynamic	(1) Explore relationships between various actors in the CE (2) Requires industrial symbiosis and social change (3) Able to model market potential (4) Able to model CE transitions at various scales (5) Industrial symbiosis captures feedback loops, which are important to industrial symbiosis (6) Able to model the CE over time	A. Data intensive B. Difficult to validate C. Difficult to generalize	(A,B) Calibration and sensitivity analysis (B) Simple, general model with further refinements	End-of-life rates, Raw Material Consumption (RMC), Waste ratio, Waste and recycling per capita, Decoupling factor, Value added at factor cost
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Do We Need a New Sustainability Assessment Method for the Circular Economy? A Critical Literature Review

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The goal of the circular economy (CE) is to transition from today's take-make-waste linear pattern of production and consumption to a circular system in which the societal value of products, materials, and resources is maximized over time. Yet circularity in and of itself does not ensure social, economic, and environmental performance (i.e., sustainability). Sustainability of CE strategies needs to be measured against their linear counterparts to identify and avoid strategies that increase circularity yet lead to unintended externalities. The state of the practice in quantitatively comparing sustainability impacts of circular to linear systems is one of experimentation with various extant methods developed in other fields and now applied here. While the proliferation of circularity metrics has received considerable attention, to-date, there is no critical review of the methods and combinations of methods that underlie those metrics and that specifically quantify sustainability impacts of circular strategies. Our critical review herein analyzes identified methods according to six criteria: temporal resolution, scope, data requirements, data granularity, capacity for measuring material efficiency potentials, and sustainability completeness. Results suggest that the industrial ecology and complex systems science fields could prove complementary when assessing the sustainability of the transition to a CE. Both fields include quantitative methods differing primarily with regard to their inclusion of temporal aspects and material efficiency potentials. Moreover, operations research methods such as multiple-criteria decision-making (MCDM) may alleviate the common contradictions which often exist between circularity metrics. This review concludes by suggesting guidelines for selecting quantitative methods most appropriate to a particular research question and making the argument that while there are a variety of existing methods, additional research is needed to combine existing methods and develop a more holistic approach for assessing sustainability impacts of CE strategies.

Keywords: circular economy, material efficiency, industrial ecology, circularity metrics, sustainability assessment, complex systems science

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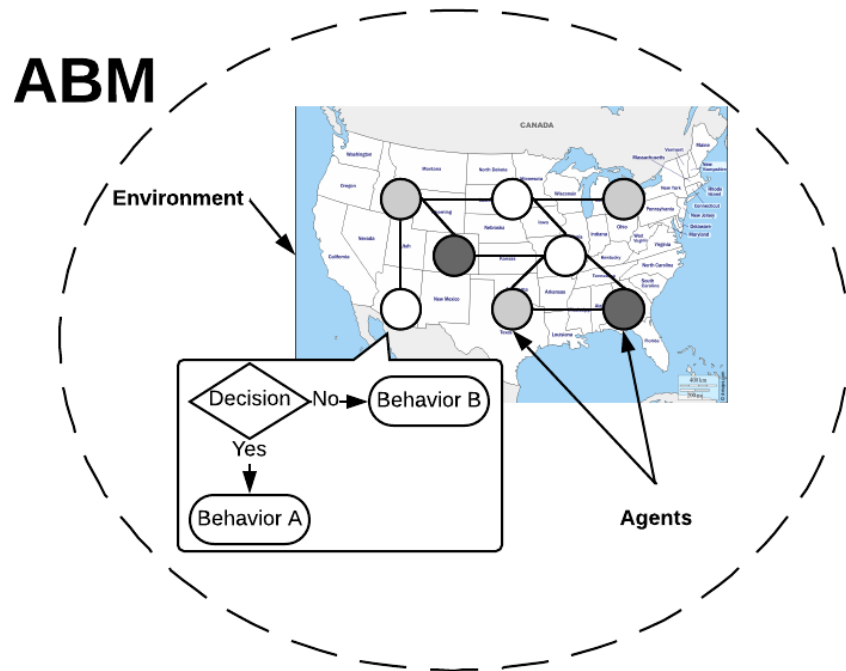
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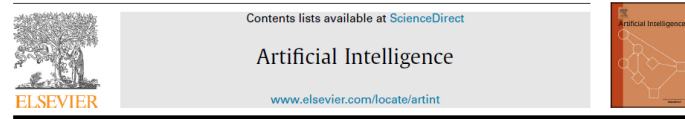
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Background – Agent-based modeling

- A good method to model human behaviors is agent-based modeling.
- Agent-based modeling (ABM):
 - Bottom-up modeling where each agent follows its own behavioral rules.
 - **Agent:** individual entity which has its own characteristics, behaviors and can interact with each other and with the environment.
 - **Goal:** Understand how a system's macro level behavior emerges from the individual behaviors of the agents.
- Advantages of the ABM method:
 - Model **individual decisions** and **peer effects**.
 - Represent a population's **heterogeneity**.



Examples – ABM are used in widely different fields



Mis- and disinformation in a bounded confidence model

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ABSTRACT

The bounded confidence model has been widely used to formally study groups of agents who are sharing opinions with those in their epistemic neighborhood. We revisit the model with an eye toward studying mis- and disinformation campaigns, which have been much in the news of late. To that end, we introduce typed agents into the model, specifically agents who can be irresponsible in different ways, most notably, by being deceitful, but also by being reluctant to try and obtain information from the world directly. We further add a mechanism of confidence dynamics to the model, which—among other things—allows agents to adapt the closeness threshold for counting others as being their epistemic neighbors. This will be used to study the effectiveness of possible defense mechanisms against mis- and disinformation efforts.

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ABM has been used in multiple fields:

- ABM of covid-19 spread in the Boston metropolitan area (Aleta et al., 2020).
- ABM of the spread of mis and disinformation (Douven & Hegselmann, 2021).
- An ABM revisiting Schelling segregation model using reinforcement learning (Sert et al., 2020).

ARTICLES

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nature
human behaviour



Modelling the impact of testing, contact tracing and household quarantine on second waves of COVID-19

Alberto Aleta¹, David Martin-Corral^{2,3}, Ana Pastore y Piontti⁴, Marco Ajelli^{5,6}, Maria Litvinova¹, Matteo Chinazzi⁴, Natalie E. Dean⁷, M. Elizabeth Halloran^{8,9}, Ira M. Longini Jr⁷, Stefano Merler⁴, Alex Pentland¹⁰, Alessandro Vespignani^{1,4}, Esteban Moro^{2,10} and Yamir Moreno^{1,12}

While severe social-distancing measures have proven effective in slowing the coronavirus disease 2019 (COVID-19) pandemic, second-wave scenarios are likely to emerge as restrictions are lifted. Here we integrate anonymized, geolocalized mobility data with census and demographic data to build a detailed agent-based model of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) transmission in the Boston metropolitan area. We find that a period of strict social distancing followed by a robust level of testing, contact-tracing and household quarantine could keep the disease within the capacity of the healthcare system while enabling the reopening of economic activities. Our results show that a response system based on enhanced testing and contact tracing can have a major role in relaxing social-distancing interventions in the absence of herd immunity against SARS-CoV-2.

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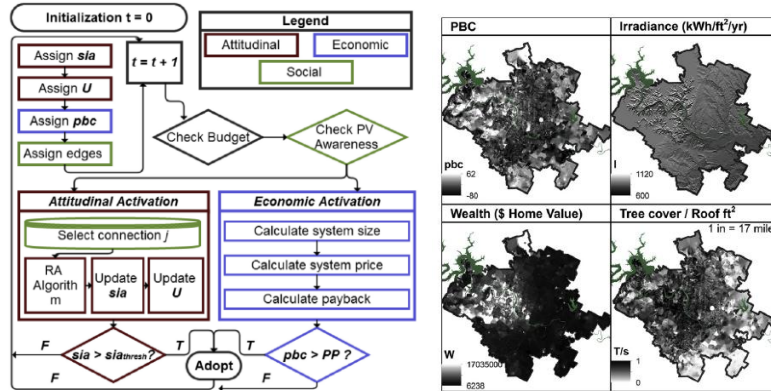
Segregation dynamics with reinforcement learning and agent based modeling

Egemen Sert^{1,2}, Yaneer Bar-Yam¹ & Alfredo J. Morales^{1,3,12}

Societies are complex. Properties of social systems can be explained by the interplay and weaving of individual actions. Rewards are key to understand people's choices and decisions. For instance, individual preferences of where to live may lead to the emergence of social segregation. In this paper, we combine Reinforcement Learning (RL) with Agent Based Modeling (ABM) in order to address the self-organizing dynamics of social segregation and explore the space of possibilities that emerge from considering different types of rewards. Our model promotes the creation of interdependencies and interactions among multiple agents of two different kinds that segregate from each other. For this purpose, agents use Deep Q-Networks to make decisions inspired on the rules of the Schelling Segregation model and rewards for interactions. Despite the segregation reward, our experiments show that spatial integration can be achieved by establishing interdependencies among agents of different kinds. They also reveal that segregated areas are more probable to host older people than diverse areas, which attract younger ones. Through this work, we show that the combination of RL and ABM can create an artificial environment for policy makers to observe potential and existing behaviors associated to rules of interactions and rewards.

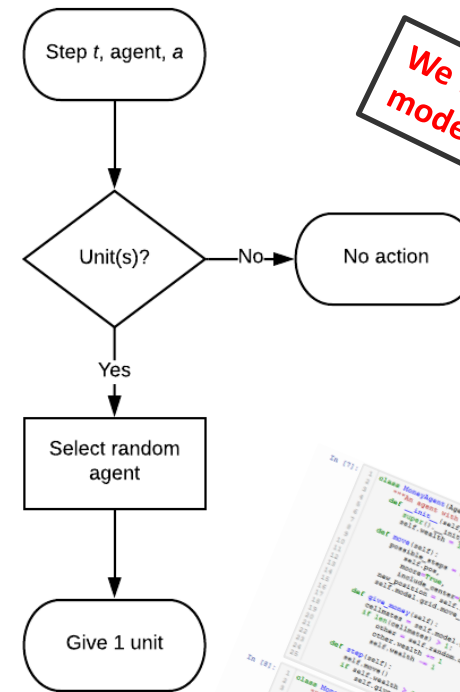
Examples – ABMs can be simple or sophisticated

- ABM can be highly sophisticated or fairly simple.



Rai & Robinson (2015), ABM of residential solar adoption
→ run on Texas Advanced
Computing Center's supercomputer

MESA's creators Boltzmann wealth model (MESA is a python
library developed in 2015) → less than 50 lines of code



We will study this
model later on!

```

In [7]:
class BoltzmannModel:
    def __init__(self, N, T, S, R, G, H, W, L, U, P, C, D, E, F, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z):
        self.N = N
        self.T = T
        self.S = S
        self.R = R
        self.G = G
        self.H = H
        self.W = W
        self.L = L
        self.U = U
        self.P = P
        self.C = C
        self.D = D
        self.E = E
        self.F = F
        self.I = I
        self.J = J
        self.K = K
        self.L = L
        self.M = M
        self.N = N
        self.O = O
        self.P = P
        self.Q = Q
        self.R = R
        self.S = S
        self.T = T
        self.U = U
        self.V = V
        self.W = W
        self.X = X
        self.Y = Y
        self.Z = Z

    def step(self):
        # Create agents
        for i in range(self.N):
            agent = BoltzmannAgent(self, i)
            self.agents.append(agent)

        # Create space
        space = BoltzmannSpace(self)
        self.space = space

        # Add agents to space
        for agent in self.agents:
            self.space.add_agent(agent)

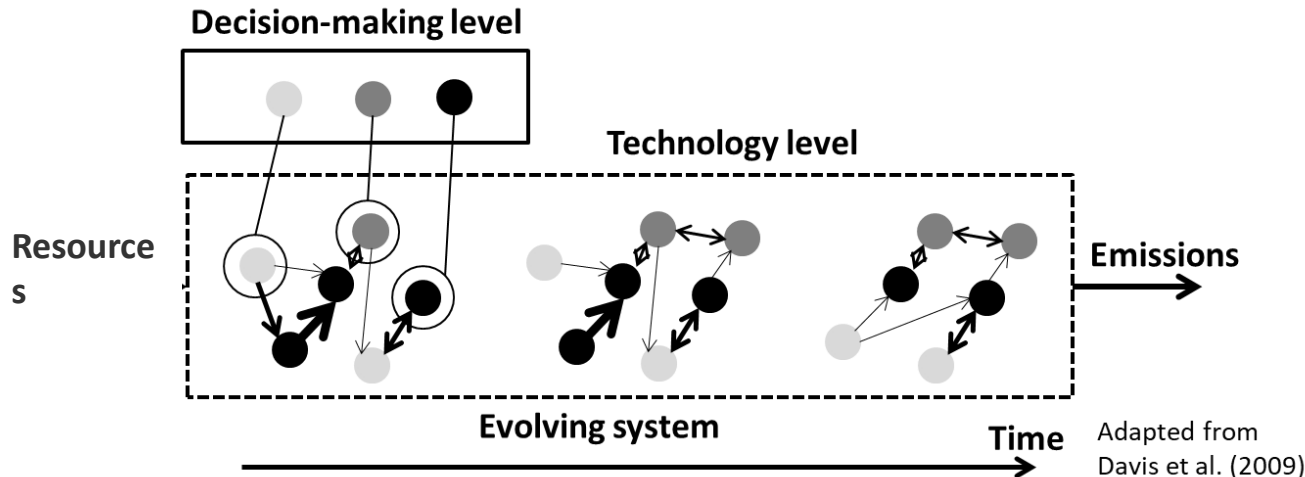
        # Run simulation
        self.run_simulation()

    def run_simulation(self):
        # Run simulation
        for i in range(1000):
            self.step()

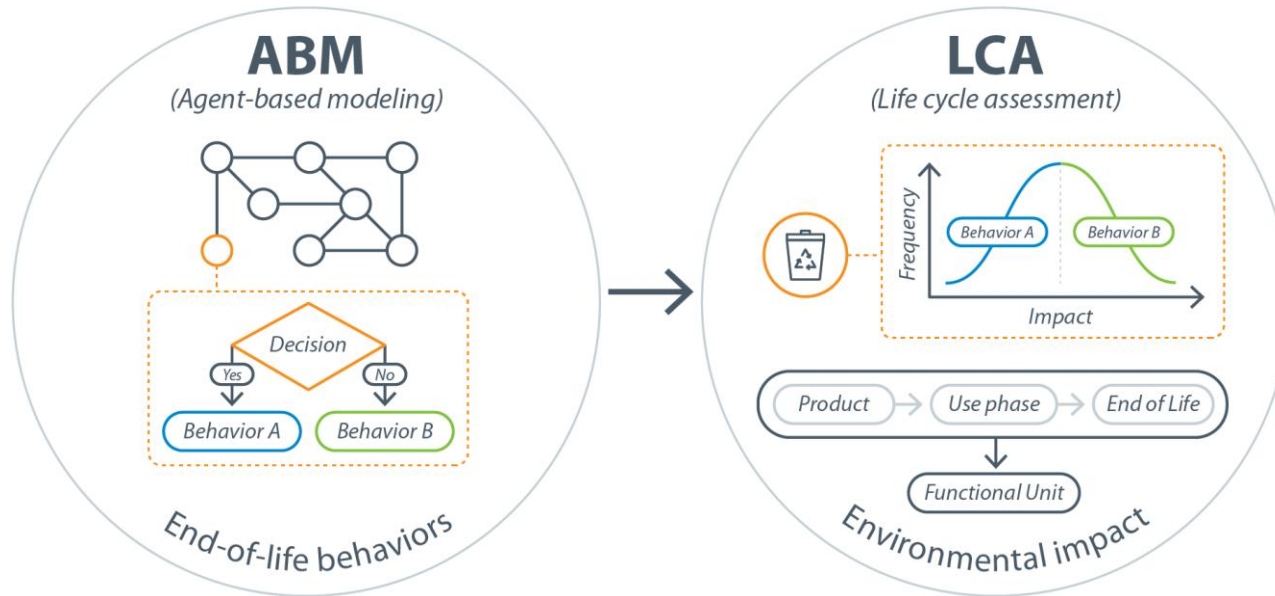
        # Print results
        self.print_results()
    
```

Examples – ABM and LCA

- ABM has been used to model complex systems in LCA:
 - LCA and ABM can be combined unidirectionally or merged.
 - Examples include study of farmers' adoption of biofuel crops, technological choice to produce electricity, LED adoption, green behavior adoption.



Examples – ABM and LCA

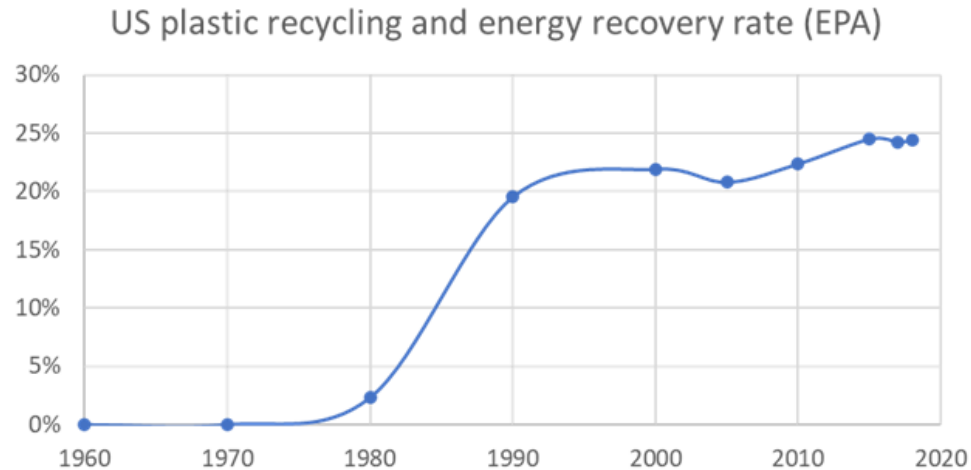


Example of the use of combined ABM and LCA in the context of the CE:

- ABM represents the decisions of various CE actors (e.g., consumers, manufacturers, recyclers), and LCA evaluates the decisions' environmental impacts.
- For instance, a typical decision for consumers would be, to recycle a product or dump it.
- ABM provides better scenario modeling capacity that might lead to more realistic LCA results (Micolier et al., 2019).

Case Study – Problem Statement

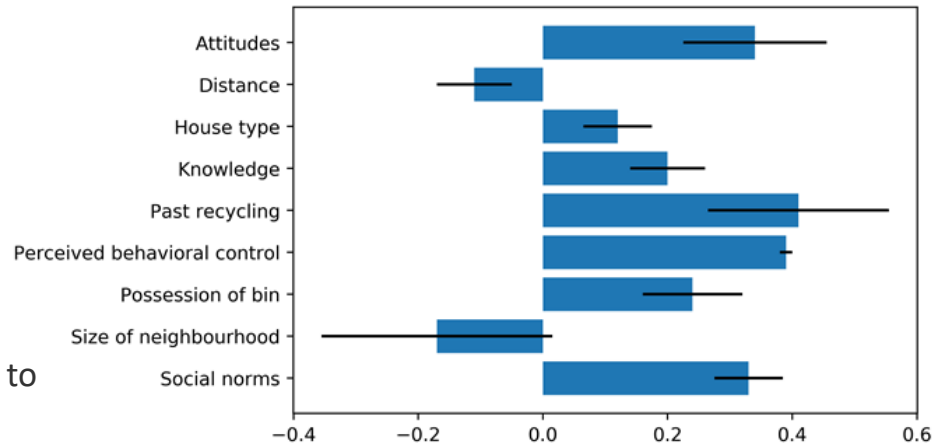
- The United States recycling rate has plateaued in the past few years; if anything, it is decreasing slightly:
 - In 2018 about 32% of the 292 million tons of municipal solid waste (MSW) was sent to recycling and composting programs.
 - The US MSW recycling rate lags behind other developed countries (e.g., Switzerland = 51%, UK = 43%).
 - Contamination issues hinder recycling due to phenomenon such as “wishcycling”.
 - Recycling behaviors are highly contextual making “one-size-fits-all” solutions sub-optimal.



<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data>
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Case Study – Many Factors Affect Recycling Behaviors

- According to Raghu & Rodrigues (2020) systematic literature review, factors can be classified in 4 broad categories:
 - Psychological factors.
 - Situational factors.
 - Community factors.
 - Demographic factors.
- Geiger et al. (2019) quantitative meta-analysis of factors affecting recycling behaviors:
 - Analyzed 91 empirical studies.
 - Psychological and situational factors can be targeted to increase recycling:
 - Reducing traveled distance (e.g., by developing curbside recycling programs).
 - Providing information feedback (e.g., via inspection and cart tagging in Olympia, WA).



Effect sizes of some of the psychological and situational factors affecting recycling identified in Geiger et al., 2019 meta-analysis – error bars = 95% confidence interval

Case Study – Behavioral Interventions Examples

Information feedback (via inspection and cart tagging) in Olympia, WA. Source: <https://resource-recycling.com>



Composting bins in Montréal, Canada.
Source: <https://www.shutterstock.com>

Case Study – Behavioral Rules

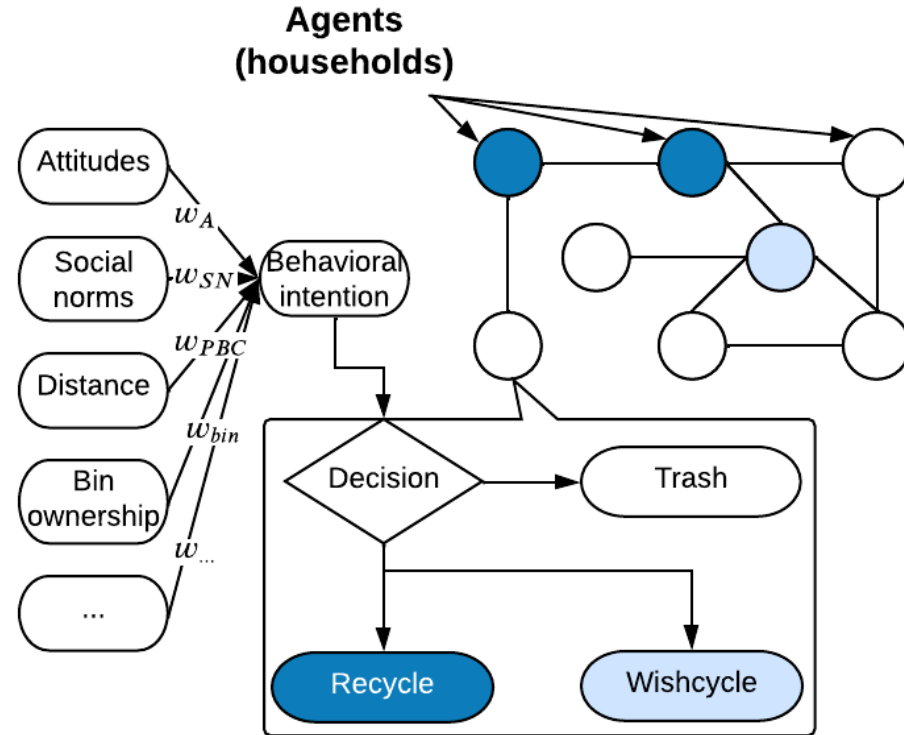
- Three behavior adoption dynamics are modeled: trash, recycling, and “wishcycling”:

- The habitual nature of the disposal behavior (i.e., trash versus wishcycling/recycling) is captured according to the Alphabet theory (Taghikhah et al., 2020).
- If the agent “ponders” his disposal behavior (i.e., not habitual), the variables m from Geiger et al. (2019) seen above are used to define the probability BI_{ij}^t that a given behavior j is adopted by agent i at time t .
- Probabilities P_{ij}^t are used to represent agents’ bounded rationality and factors not considered in the ABM (e.g., attitude-behavior gap).

$$P_{ij}^t = \begin{cases} 1 & \text{for past behavior } j \text{ if habit conditions are met} \\ BI_{ij}^t & \text{otherwise} \end{cases}$$

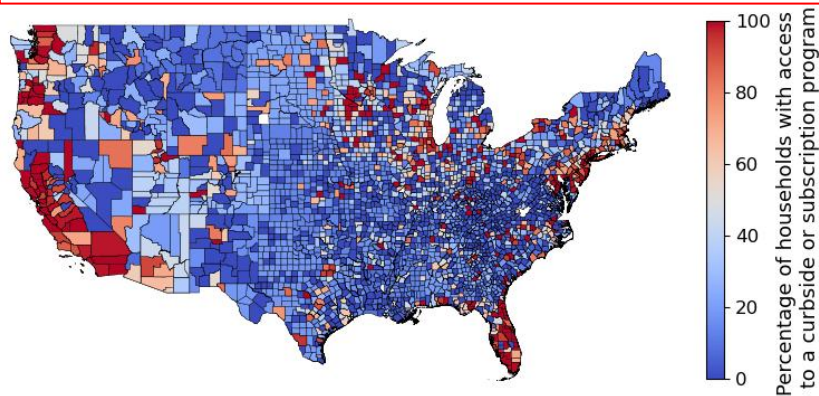
$$BI_{ij}^t = \sum_m w_m m_{ij}^t$$

With weights w_m taken from Geiger et al., 2019

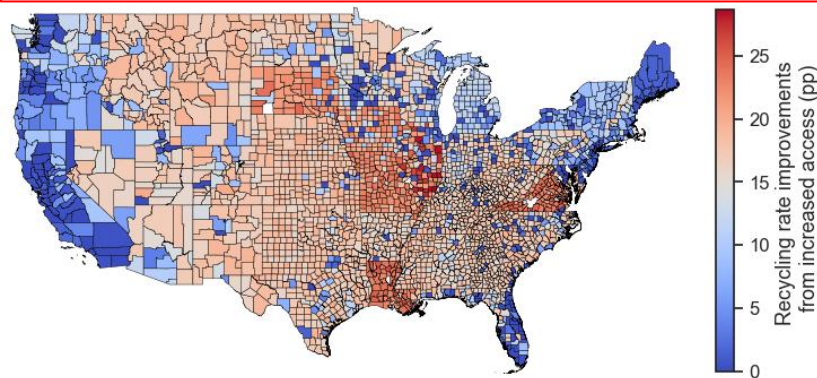


Case Study – Investigating the Effects of Increased Access to Recycling Programs

Baseline access to (“easy”) curbside or multifamily recycling programs



PET recycling improvements when 100% of households without easy access are targeted



- Generally, counties with a low access initially see the highest improvements in recycling rates due to the “improved access” intervention (e.g., in Louisiana and Virginia)
- However, there are nuances: states with low access to curbside recycling but with bottle bills (e.g., Maine) already have high collection rates → increasing access to recycling programs has a lower effect.
- Prioritizing investments in certain counties could more efficiently increase the amount of recycled PET bottles.
- Improved access is only one factor affecting recycling behaviors; other factors such as recycling knowledge, habits, and social norms can also be targeted to improve recycling rates.

Case Study – Conclusions

- The ABM can be combined to 4P ([Ghosh et al. 2023](#)) to calculate the environmental and circularity benefits of recycling. Both intervention scenarios increase the overall recycling rate by about 20 percentage point, leading to:
 - Reduction in emissions of 0.2 and 0.3 million metric tons per year for mechanical and glycolysis recycling of PET, respectively.
 - Circularity gains (as measured by the material circularity indicator) of 3% and 7% for mechanical and glycolysis recycling of PET, respectively.
- The ABM was also used by Argonne National Laboratory's RELOG model to design optimized recycling infrastructure and evaluate their costs under different intervention scenarios.
- Future work could study the effect of combined interventions (e.g., increased education and improved access) and expand the model to include more materials (e.g., aluminum cans or organic waste).

Let's practice!!

https://colab.research.google.com/drive/1-F-z7ryuONBsyv_cBJNS1jke_tK0uUhQ#scrollTo=tRe2V82IBxfN



AB-LCA - Google Collab

(Repository)



<https://github.com/jwalzberg/ACLCA-Workshop-AB-LCA/tree/ISSST-2026>

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